



# Real Estate Market Analysis With Power BI

## Group Name: Estate Insights

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## 1. Introduction

The real estate market is an important sector that supports economic growth and urban development. Real estate data usually includes connected information such as properties, customers, agents, listings, and transactions. Since this data is stored in different tables, it can be difficult to analyze manually and understand the full market performance.

This project, Real Estate Market Analysis with Power BI, aims to create an interactive dashboard that presents real estate data in a clear and simple way. The dashboard uses Power BI to transform raw data into useful KPIs, charts, filters, and insights.

The dashboard focuses on the main areas shown in the visuals, including total revenue, total transactions, property types, customer distribution, acquisition channels, agent performance, market speed, and future revenue forecast. It supports descriptive analysis by summarizing market performance, diagnostic analysis by showing factors related to revenue and deal price, and predictive analysis by forecasting future revenue.

The project is also related to SDG Goal 8: Decent Work and Economic Growth and SDG Goal 9: Industry, Innovation, and Infrastructure. It supports better business decisions in the real estate sector and shows how digital tools such as Power BI can improve data analysis and reporting.

The main goal of this project is to provide a user-friendly business intelligence dashboard that helps real estate stakeholders understand the market and make better decisions based on the dashboard results.

## 2. Roles and contribution of team members

| Student Name      | Student ID | Contribution                                                       |
|-------------------|------------|--------------------------------------------------------------------|
| Sara Al-Qahtani   | 445008574  | Data cleaning, dashboard formatting, measures, and report editing. |
| Lamar Bahaydan    | 445008616  | Overview dashboard page and measures.                              |
| Dana Almkhlafi    | 445008596  | Agents analysis page and measures.                                 |
| Alshaima AlOtaibi | 445008542  | Customer analysis page, measures, and report editing.              |
| Salma Algedeiri   | 445008552  | Predictive analysis page and measures.                             |

*Table 1 Roles and contribution of team members*



### 3. Problem description

Stakeholders in the real estate market, such as managers, agents, and investors, need clear information to understand market performance and make better decisions. However, the data is stored in separate tables, such as customers, properties, agents, listings, and transactions. This makes it harder to connect the data and see the full picture.

The main problem is the lack of one interactive dashboard that can summarize the market and answer important questions, such as:

Which cities have the highest number of transactions?

Which property types are most common?

How much total revenue was generated?

How long do properties usually stay on the market?

Which customer acquisition channels bring more customers?

How do agents perform based on city, experience, rating, and revenue?

What is the expected revenue trend for the next months?

Without a dashboard, users may need to check several tables manually. This takes time and may lead to slower decisions. Therefore, this project solves the problem by building a Power BI dashboard that combines the data into one interactive and organized view.

### 4. Overview of solution

The solution is an interactive Power BI dashboard designed to analyze the real estate market from different sides. The dashboard connects five main tables: customers, agents, properties, listings, and transactions. These tables are linked using IDs such as `customer_id`, `agent_id`, `property_id`, and `listing_id`. This matches the project requirement of using more than one table.

The dashboard includes four main pages:

#### Overview Dashboard

The Overview Dashboard gives a general summary of the real estate market. It includes KPI cards such as Total Revenue, Total Transactions, Total Properties, Total Customers, Average Deal Price, and Average Days on Market.

It also includes visuals for revenue trend over time, transactions by city, property type distribution, and market speed category. These visuals help users understand the overall market performance quickly.

#### Customers Analysis Dashboard

The Customers Analysis page focuses on customer distribution and acquisition channels. It includes city and acquisition channel filters to help users explore the data more easily. The page shows customers by city, customer segment, and acquisition channel. This helps users understand where customers come from and which customer groups are more common.

#### Agent Performance Dashboard

The Agent Performance page compares agents based on city, revenue, experience, and rating. It includes KPI cards and visuals such as total revenue by agent city and agent experience versus agent rating.



This page helps management compare agent performance and identify stronger locations or areas that may need improvement.

### **Predictive Analytics Dashboard**

The Predictive Analytics page focuses on future revenue and factors related to deal price and revenue. It includes top revenue factors, future revenue forecast, Key Influencers, and Decomposition Tree visuals.

The forecast visual helps users understand the expected revenue trend for the next months. The Key Influencers visual helps identify factors related to average deal price.

The Decomposition Tree breaks down total revenue by categories such as city and property type. Together, these visuals help users understand where revenue comes from and what factors may affect performance.

Overall, the dashboard provides a simple and organized way to explore real estate data. It helps users understand revenue, transactions, customer behavior, agent performance, property types, selling speed, and future revenue trends using the visuals included in the Power BI dashboard.

## **5. Description of data**

The data used in this project was sourced from a comprehensive Real Estate Analytics environment available on Kaggle. This dataset was selected because it provides a rich, multi-table structure that reflects how modern real estate platforms and brokerages organize their analytical data. It includes over 1,000 records, ensuring it meets the project's technical requirements for robust analysis.

The dataset is organized into five primary tables, each capturing a specific dimension of the real estate business:

### **5.1 Customers Table**

The customers table contains customer profile and acquisition data. It includes information about customers' demographics, segmentation, acquisition channels, and behavioral indicators. This table helps analyze customer groups and understand how different customer characteristics may relate to real estate transactions.

The main columns in this table are:

- customer\_id
- signup\_date
- home\_city
- segment
- age
- household\_size
- income\_band
- acquisition\_channel
- propensity\_score



## 5.2 Agents Table

The agents table contains information about real estate agents and brokers. It includes details about the agent's city, years of experience, base commission rate, and rating. This table is useful for analyzing agent performance and comparing agents based on experience and rating.

The main columns in this table are:

- agent\_id
- agent\_city
- experience\_years
- base\_commission\_rate
- agent\_rating

## 5.3 Properties Table

The properties table contains property inventory and property characteristics. It includes information about property location, type, size, number of bedrooms and bathrooms, year built, location score, amenities, and listing price. This table helps analyze which property features may affect deal prices and property sales performance.

The main columns in this table are:

- property\_id
- city
- neighborhood\_code
- property\_type
- size\_sqm
- bedrooms
- bathrooms
- year\_built
- location\_score
- amenities\_count
- list\_price
- has\_parking
- near\_transit
- near\_school



#### **5.4 Listings Table**

The listings table contains property listing lifecycle data. It includes information about when a property was listed, the listing channel, listed price, sale status, days on market, and closing date. This table is important for the diagnostic analysis because it helps study why some properties sell faster while others stay longer on the market.

The main columns in this table are:

- listing\_id
- property\_id
- agent\_id
- listing\_date
- listing\_channel
- listed\_price
- sold\_flag
- days\_on\_market
- close\_date

#### **5.5 Transactions Table**

The transactions table contains closed deal and revenue data. It includes information about transaction ID, listing ID, property ID, agent ID, customer ID, city, deal date, deal price, commission amount, payment mode, and deal status. This table is important for financial analysis and forecasting because it is used to calculate total revenue, average deal price, total commission, and monthly revenue trends.

The main columns in this table are:

- transaction\_id
- listing\_id
- property\_id
- agent\_id
- customer\_id
- city
- deal\_date
- deal\_price
- commission\_amount
- payment\_mode
- deal\_status



## 5.6 Importance of the Data

Overall, the dataset provides a strong foundation for building a real estate analytics dashboard. The customers table helps understand buyer profiles, the agents table supports agent performance analysis, the properties table explains property characteristics, the listings table shows market activity and selling speed, and the transactions table provides revenue and commission information.

By combining these tables in Power BI, the dashboard can provide useful insights for real estate agents and management. It helps identify high-value deals, compare property performance, analyze listing delays, evaluate agent performance, and forecast future revenue.

## 6. Data Preparation.

Data preparation was performed using Power Query Editor in Power BI. Since the dataset was already mostly clean and the data types were correct, the preparation focused on organizing the data, removing only the missing values found in important columns, and creating analytical columns that support the dashboard objectives.

The main goal of this step was to prepare the data for both Predictive Analytics and Diagnostic Analytics. The dashboard focuses on analyzing real estate revenue, property selling speed, agent performance, listing activity, and commission performance. Therefore, the transformations were designed to support these areas directly.

### 6.1 Agents Table Preparation

For the agents table, the data was imported into Power BI and prepared using Power Query Editor. The first row was promoted as headers, and the data types were checked and corrected based on each column.

A filtering step was applied to the agent\_rating column to remove null or empty values. This step was important because agent rating is used in the dashboard to analyze and compare agent performance.

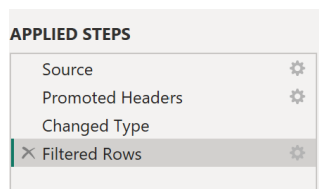


Figure 1 Applied steps for agents table in Power Query Editor.



## 6.2 Listings Table Preparation

For the listings table, the data was prepared to support the diagnostic analysis of property selling speed. This table includes important fields such as listing\_id, property\_id, agent\_id, listing\_date, listing\_channel, listed\_price, sold\_flag, days\_on\_market, and close\_date.

A custom column named Market Speed Category was created based on the days\_on\_market column. This column classifies property listings into three categories: Fast Sale, Normal Sale, and Delayed Sale. This classification helps identify which properties were sold quickly and which properties stayed longer on the market. The Market Speed Category supports the diagnostic part of the dashboard because it helps analyze the factors that may affect selling speed, such as city, listing channel, price, and agent.

A Start of Month column was also created from the listing\_date column to support monthly listing trend analysis. After that, the columns were reordered and renamed to make the table clearer and easier to use in the dashboard.

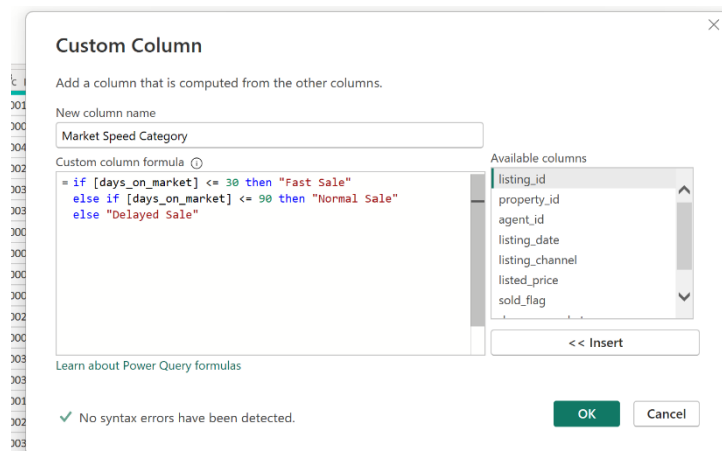


Figure 2 Market Speed Category custom column.

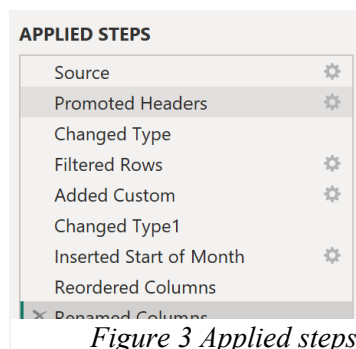


Figure 3 Applied steps for listings table in Power Query Editor.



### 6.3 Transactions Table Preparation

For the transactions table, the data was prepared to support predictive and financial analysis. This table includes transaction details such as transaction\_id, listing\_id, property\_id, agent\_id, customer\_id, city, deal\_date, deal\_price, commission\_amount, payment\_mode, and deal\_status.

Rows with missing important values were filtered to improve the accuracy of the financial analysis. A Deal Month column was created from the deal\_date column. This helped organize transactions by month and supported revenue trend analysis and forecasting in Power BI.

A custom column named Revenue Category was created based on the deal\_price column. Since the deal prices in the dataset were high, the revenue categories were adjusted to match the actual data range. Transactions were classified into Low Revenue, Medium Revenue, and High Revenue groups. This helps compare transactions based on their financial value.

Another custom column named Commission Rate was created using commission\_amount and deal\_price. The data type of this column was changed to Percentage to make commission values easier to understand and compare.

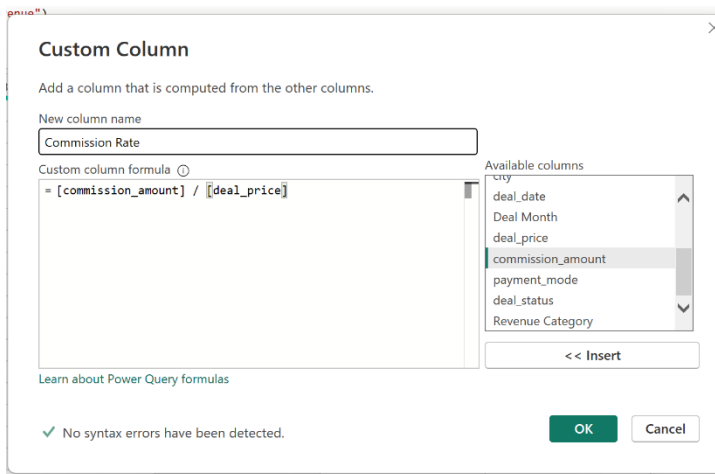


Figure 4 Commission Rate custom column

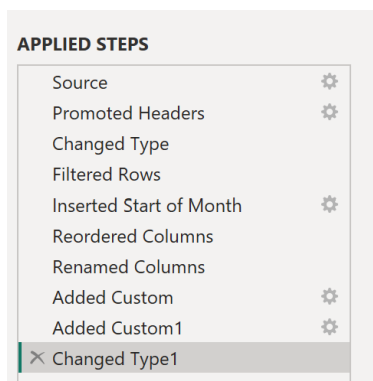


Figure 5 Applied steps for transactions table in Power Query Editor



#### 6.4 Custom Columns Created.

Several custom columns were created to make the data more useful for analysis. These columns were created in Power Query Editor and used later in dashboard visuals and filters.

| Custom Column                | Table        | Purpose                                                                                    |
|------------------------------|--------------|--------------------------------------------------------------------------------------------|
| <b>Market Speed Category</b> | Listings     | Classifies listings into Fast Sale, Normal Sale, and Delayed Sale based on days on market. |
| <b>Listing Month</b>         | Listings     | Supports monthly listing trend analysis.                                                   |
| <b>Deal Month</b>            | Transactions | Supports monthly revenue trend analysis and forecasting.                                   |
| <b>Revenue Category</b>      | Transactions | Classifies deals into Low, Medium, and High revenue groups.                                |
| <b>Commission Rate</b>       | Transactions | Calculates commission percentage compared to deal price.                                   |

*Table 2 Custom Columns Created*

#### 6.5 Summary of Data Preparation

After completing the data preparation steps, the dataset became more organized and ready to be used in the Power BI data model. The preparation process focused on the transformations that were actually needed for the dashboard, such as filtering missing values in important columns, creating useful analytical columns, renaming columns, and reordering the data structure.

The main prepared fields, such as **Market Speed Category**, **Listing Month**, **Deal Month**, **Revenue Category**, and **Commission Rate**, helped make the data easier to analyze and visualize. These transformations support the dashboard objective by helping users study property selling speed, revenue patterns, transaction behavior, and listing performance.

Overall, the data preparation phase improved the clarity and usability of the data without applying unnecessary cleaning steps, since most of the original dataset was already clean and correctly formatted.

## 7. Visual Model

The visual model was created in Power BI to organize the real estate dataset and connect related tables. The model includes five main tables: customers, agents, properties, listings, and transactions, linked using keys like customer\_id, agent\_id, property\_id, and listing\_id.

The transactions table is the main fact table because it contains key business data such as deal price, commission, deal date, payment mode, and deal status. It connects with the customers and agents tables to analyze customer activity and agent performance.

The listings table stores details like listing date, listed price, channel, sold status, and days on market. It connects with the properties table through property\_id and with the transactions table through listing\_id, helping track property performance and sales.

The properties table contains information such as city, property type, size, bedrooms, bathrooms, amenities, and price. The agents table supports performance analysis with agent ratings, experience, and commission rates, while the customers table provides customer profile details like age, income band, and acquisition channel.

This visual model makes the data more organized and easier to analyze. The relationships between tables support dashboards, filters, KPIs, and visualizations, helping users explore transactions, property performance, agent performance, customer behavior, and market trends clearly.

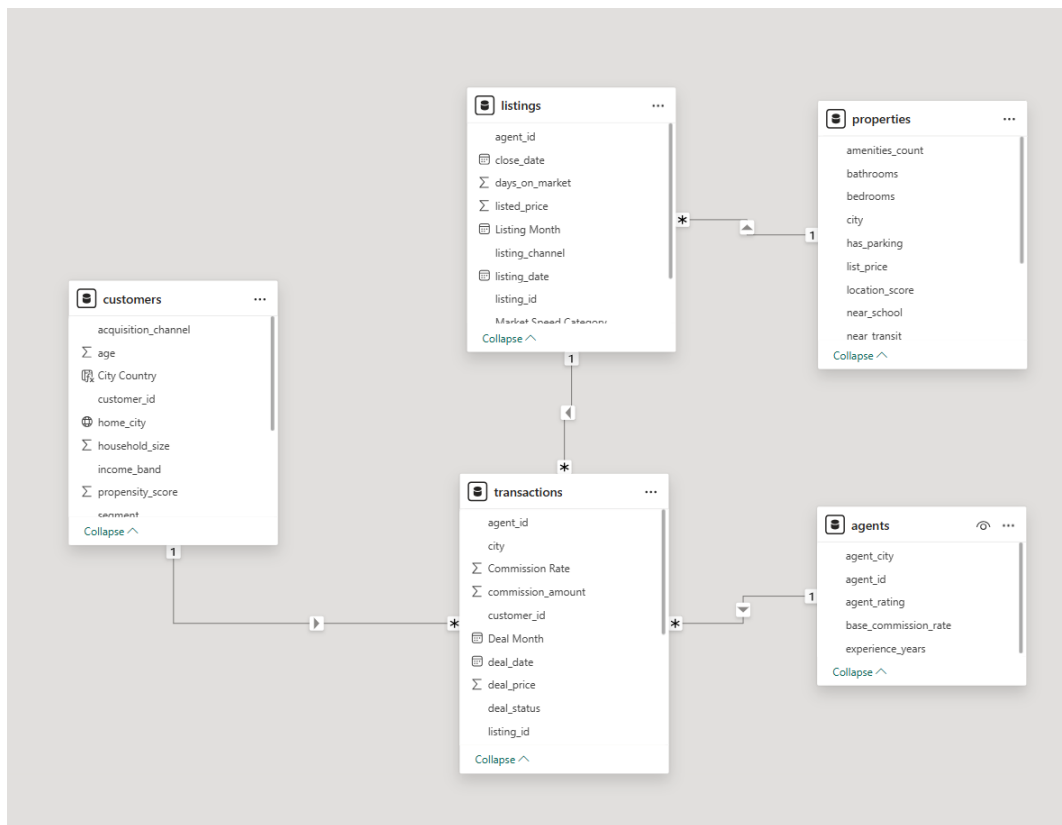


Figure 6 Visual Model

## 8. Dashboard design overview

The dashboard was designed in Power BI to analyze the real estate market in a clear and interactive way. The main purpose of the dashboard is to help users understand business performance, customer behavior, agent performance, property selling speed, and future revenue trends.

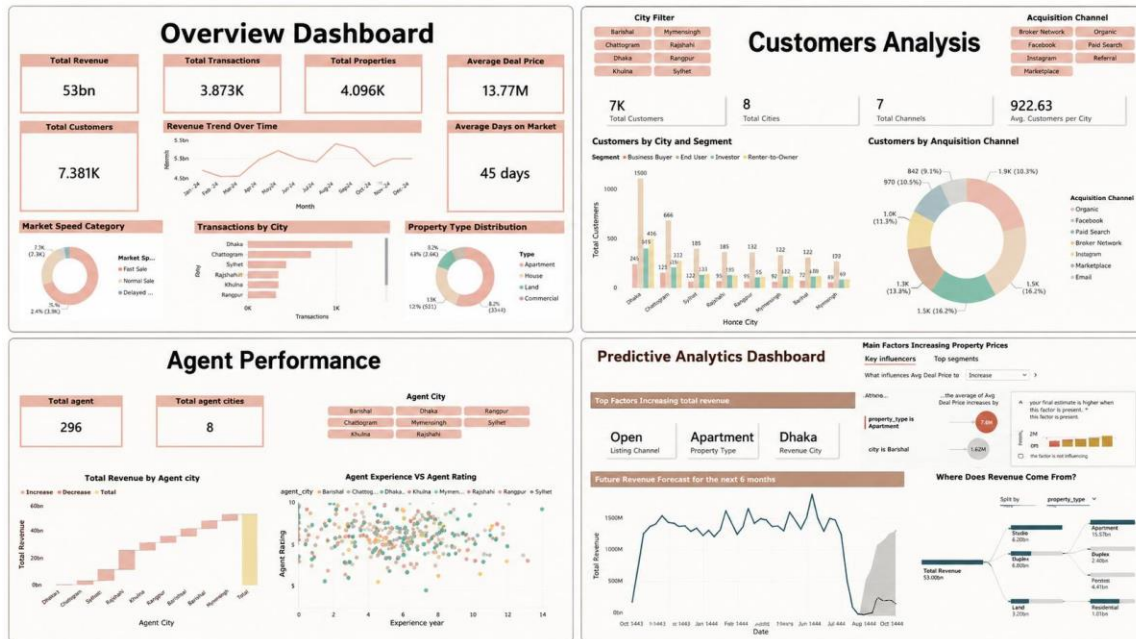


Figure 7 7 Real Estate Market Analysis Dashboard Overview

The dashboard contains four main pages: Overview Dashboard, Customers Analysis, Agent Performance, and Predictive Analytics Dashboard. Each page focuses on a specific part of the real estate business and presents the information using suitable visuals.

As shown in Figure 7, the dashboard follows a consistent design style across all pages. A light background with soft pink and neutral colors was used to create a clean and professional appearance. KPI cards were placed at the top of the pages to show the most important values, while charts, slicers, and analytical visuals were placed below to support deeper analysis.

The **Overview Dashboard** provides a general summary of the real estate market. It includes important indicators such as total revenue, total transactions, total properties, total customers, average deal price, and average days on market. It also presents visuals for revenue trend, market speed category, transactions by city, and property type distribution.



The **Customers Analysis** page focuses on customer distribution and acquisition channels. It includes filters, customer KPIs, and visuals that compare customers by city, segment, and acquisition channel.

The **Agent Performance** page evaluates real estate agents based on location, revenue, experience, and rating. It includes KPI cards and visuals that help compare agent performance across different cities.

The **Predictive Analytics** Dashboard focuses on future revenue and factors related to revenue and deal price. It includes forecasting, Key Influencers, top revenue factors, and a Decomposition Tree.

Overall, the dashboard layout was designed to be simple, readable, and easy to use. It allows users to move from a general overview to more detailed pages and supports better decision-making based on the dashboard results.

## 9. Implementation

The implementation phase focused on building the Power BI dashboard pages, creating visuals, applying filters, and using custom calculations to support real estate analysis. The dashboard was implemented using the prepared data model, which includes customers, agents, properties, listings, and transactions tables.

Different visuals were selected based on the purpose of each page. These visuals include KPI cards, line charts, donut charts, bar charts, waterfall charts, scatter plots, slicers, forecasting, Key Influencers, and Decomposition Tree visuals. These elements helped present the data clearly and supported descriptive, diagnostic, and predictive analysis.

## 9.1 Overview Dashboard Implementation

The Overview Dashboard was created to provide a general summary of the real estate business performance. This page gives users a quick understanding of the overall market situation before moving to detailed analysis pages.

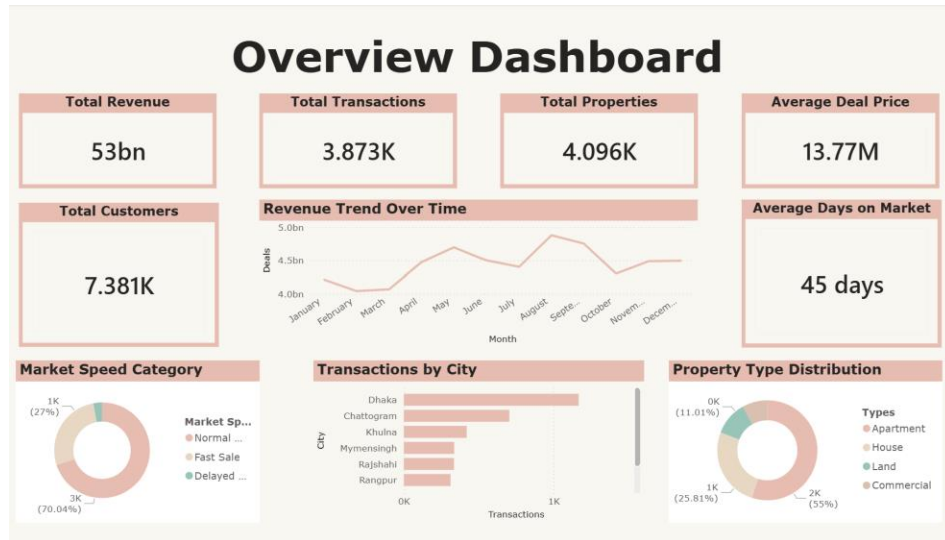


Figure 8 Power BI Dashboard Pages Overview

### 9.1.1 KPI Cards

Several KPI cards were added to show the main business indicators, including Total Revenue, Total Transactions, Total Properties, Total Customers, Average Deal Price, and Average Days on Market.

These cards help users quickly understand the overall performance of the real estate business. For example, total revenue shows the financial value of the transactions, while average days on market shows how long properties usually stay before being sold.

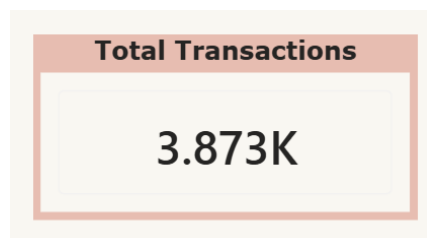


Figure 9 Example KPI in Overview

### 9.1.2 Revenue Trend Over Time

A line chart was used to show the revenue trend across months. The x-axis represents the month, while the y-axis represents total revenue.

This chart helps identify changes in revenue over time and shows which months had higher or lower revenue performance. It also supports trend analysis by giving a clear view of revenue movement.

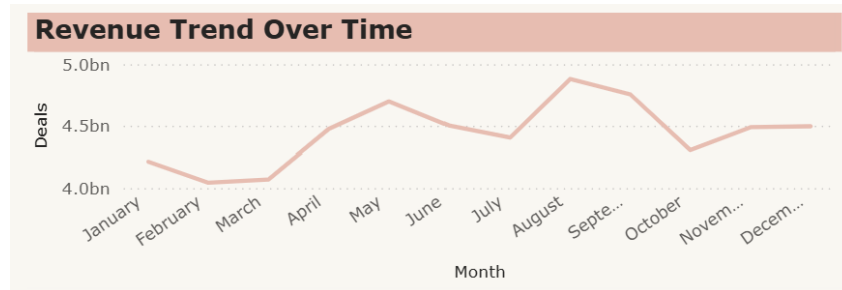


Figure 10 Revenue Trend Over Time in Overview

### 9.1.3 Market Speed Category

A donut chart was used to show the distribution of listings based on Market Speed Category. The categories include Fast Sale, Normal Sale, and Delayed Sale.

This visual helps users understand how quickly properties are sold. It supports diagnostic analysis because it shows whether most properties are selling normally or if there are delayed listings that may need further investigation.

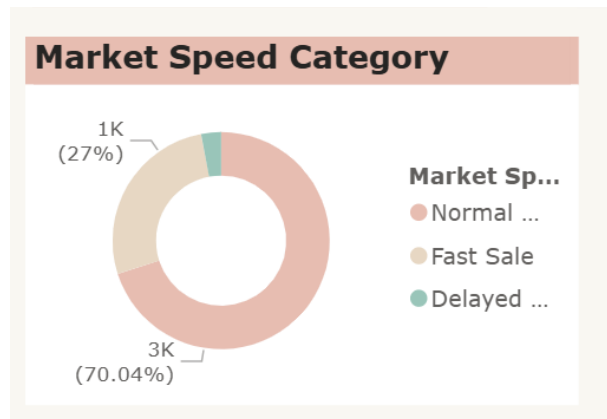


Figure 11 Market Speed Category in Overview

#### 9.1.4 Transactions by City

A bar chart was created to compare transactions across cities. This visual helps identify which cities have the highest real estate transaction activity.

By comparing cities, management can understand which locations are stronger in terms of sales activity and where business efforts may need to be improved.

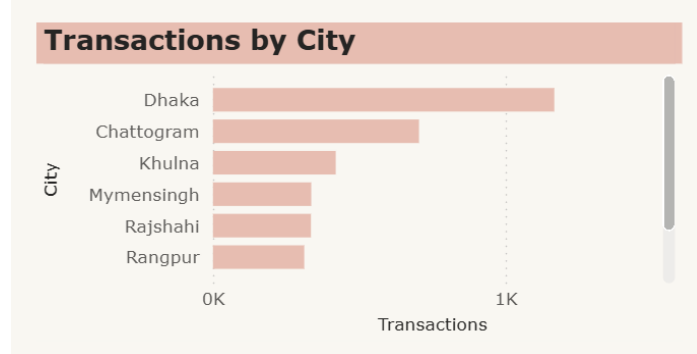


Figure 12 Transactions by City in Overview

#### 9.1.5 Property Type Distribution

A donut chart was used to show the distribution of property types, such as Apartment, House, Land, and Commercial.

This visual helps users understand which property types are most common in the dataset. It also supports business decisions by showing the dominant property categories in the market.

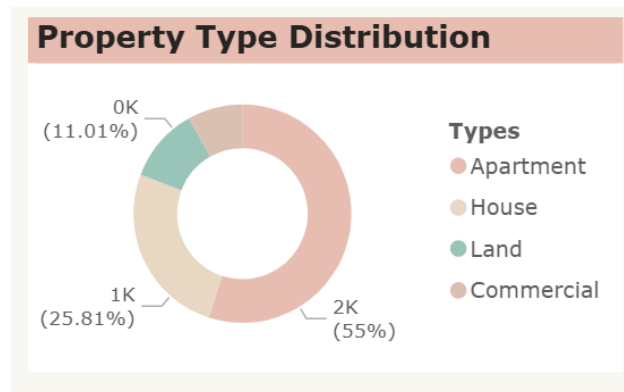


Figure 13 Property Type Distribution in Overview

## 9.2 Customers Analysis Page Implementation

The **Customers Analysis** page was created to understand customer distribution, customer segments, and acquisition channels. This page helps support marketing and customer targeting decisions.

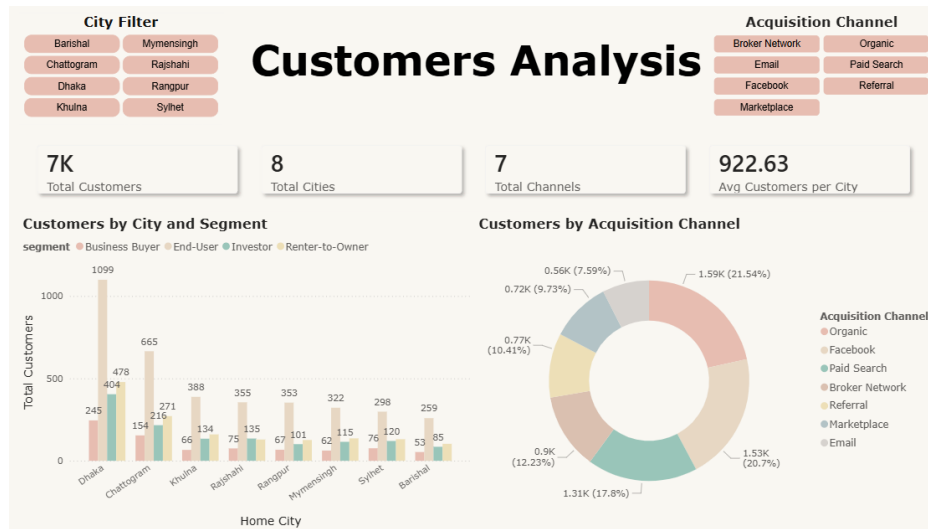


Figure 14 Customers Analysis Dashboard

### 9.2.1 Customer KPI Cards

KPI cards were added to summarize customer-related indicators, including **Total Customers**, **Total Cities**, **Total Channels**, and **Average Customers per City**. These indicators provide a quick overview of the customer base and help users understand the size and distribution of customers across cities and acquisition channels.

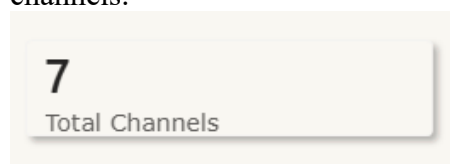


Figure 15 Example KPI in Customer



### 9.2.2 City Filter

A city slicer was added to allow users to filter the customer analysis by city. This makes the dashboard more interactive and helps users focus on specific locations. Using this filter, users can compare customer behavior in different cities and identify which city has the strongest customer presence.

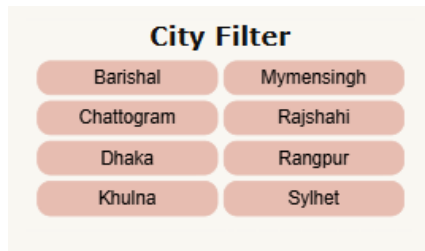


Figure 16 City Filter in Customer

### 9.2.3 Acquisition Channel Filter

An acquisition channel slicer was added to filter customers based on the channel through which they were acquired, such as **Organic, Facebook, Paid Search, Broker Network, Referral, Marketplace, and Email.** This filter helps users analyze the effectiveness of different marketing channels and understand where customers are coming from.



Figure 17 Acquisition Channel in Customer

### 9.2.4 Customers by City and Segment

A clustered column chart was used to show customers by city and segment. The segments include customer groups such as **Business Buyer**, **End-User**, **Investor**, and **Renter-to-Owner**.

This chart helps identify which customer segments are more common in each city. It supports marketing decisions because agents can adjust their strategy based on the dominant customer segment in each location.

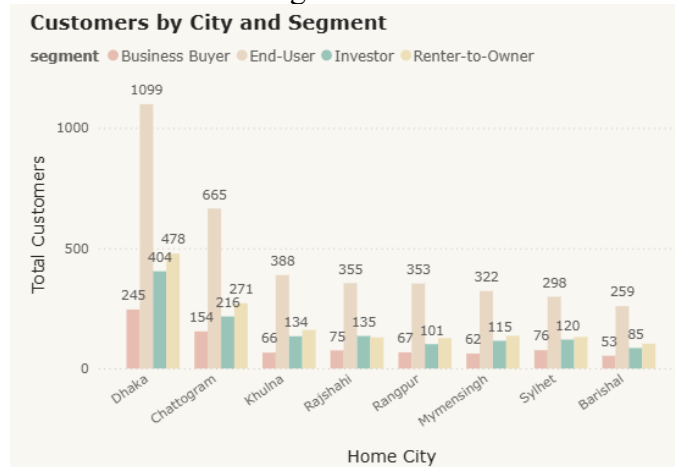


Figure 18 Customers by City and Segment

### 9.2.5 Customers by Acquisition Channel

A donut chart was created to show customer distribution by acquisition channel. This visual helps identify the channels that bring the highest number of customers. It supports marketing planning by showing which acquisition channels are more effective in attracting potential buyers.

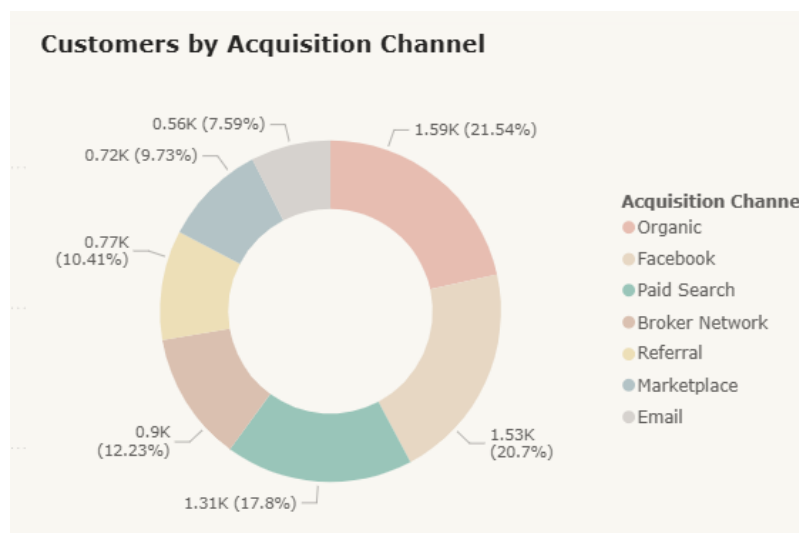


Figure 19 Customers by Acquisition Channel



### 9.3 Agent Performance Page Implementation

The **Agent Performance** page was created to evaluate real estate agents based on city, revenue, experience, and rating. This page supports management in understanding agent performance and comparing agent results across locations.

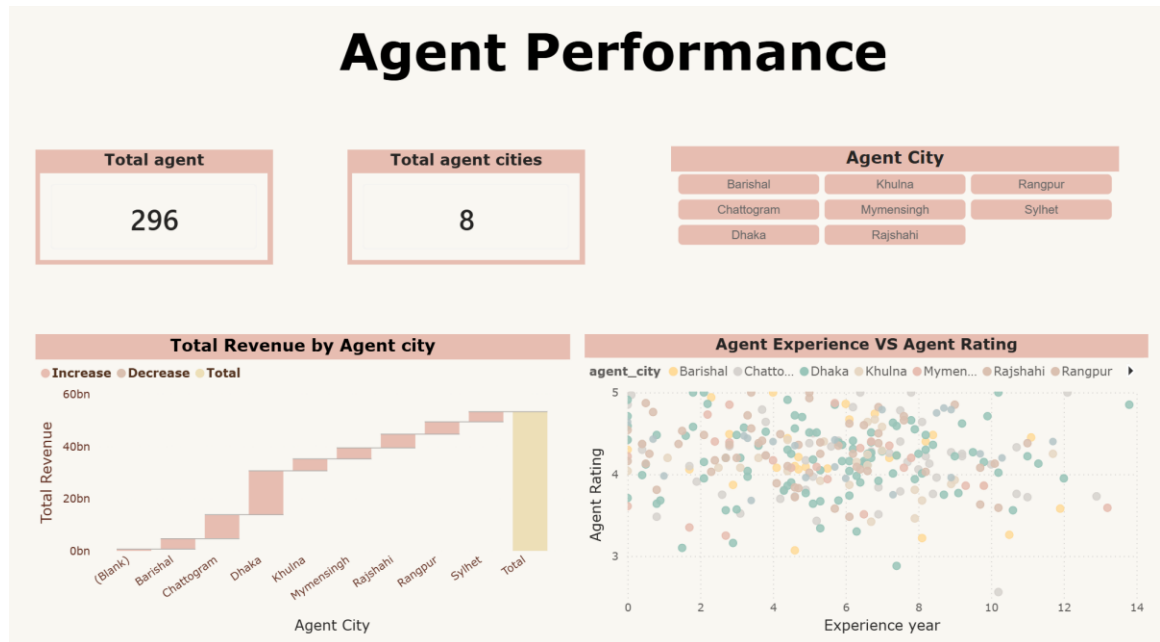


Figure 20 Agent Performance Dashboard

#### 9.3.1 Agent KPI Cards

KPI cards were added to show **Total Agents** and **Total Agent Cities**.

These cards provide a quick summary of the agent dataset and show how many agents and cities are included in the analysis.



Figure 21 Example KPI in Agent

### 9.3.2 Agent City Filter

An agent city slicer was added to allow users to filter the page based on agent location.

This filter improves interaction by allowing users to focus on specific cities and compare agent performance based on location.



Figure 22 Agent City Filter

### 9.3.3 Total Revenue by Agent City

A waterfall chart was used to show total revenue by agent city.

This visual explains how each city contributes to total revenue. It helps identify which agent cities generate the highest revenue and supports performance comparison between different locations.

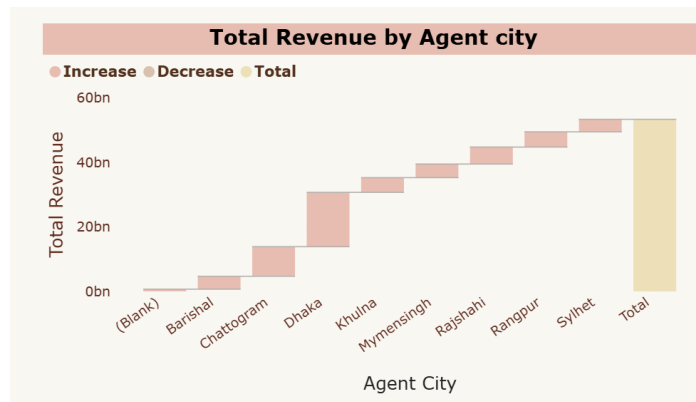


Figure 23 Total Revenue by Agent City

### 9.3.4 Agent Experience vs Agent Rating

A scatter plot was used to compare **agent experience years** with **agent rating**. The x-axis represents experience years, while the y-axis represents agent rating. This visual helps analyze whether agents with more experience tend to have higher ratings. It also helps identify agents with strong performance based on both experience and rating.

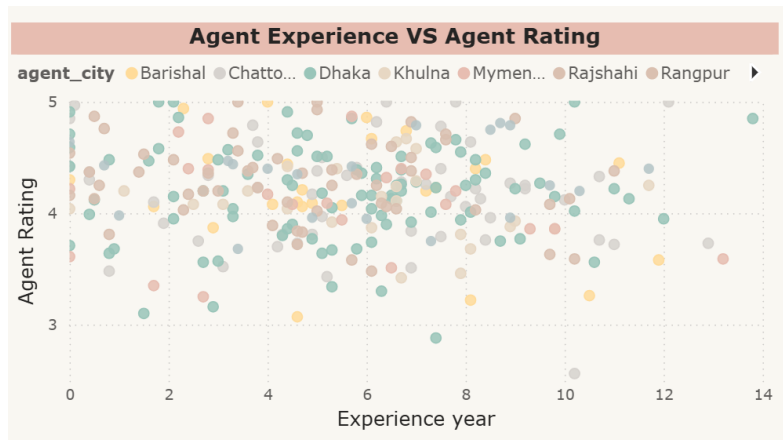


Figure 24 Agent Experience vs Agent Rating



## 9.4 Predictive Analytics Dashboard Implementation

The Predictive Analytics Dashboard was created to support future revenue planning and identify the main factors related to revenue and deal price. This page includes top revenue factors, future revenue forecast, Key Influencers, and Decomposition Tree visuals.

These visuals help users understand expected revenue trends, explore what affects average deal price, and identify which city, property type, or listing channel contributes more to revenue.

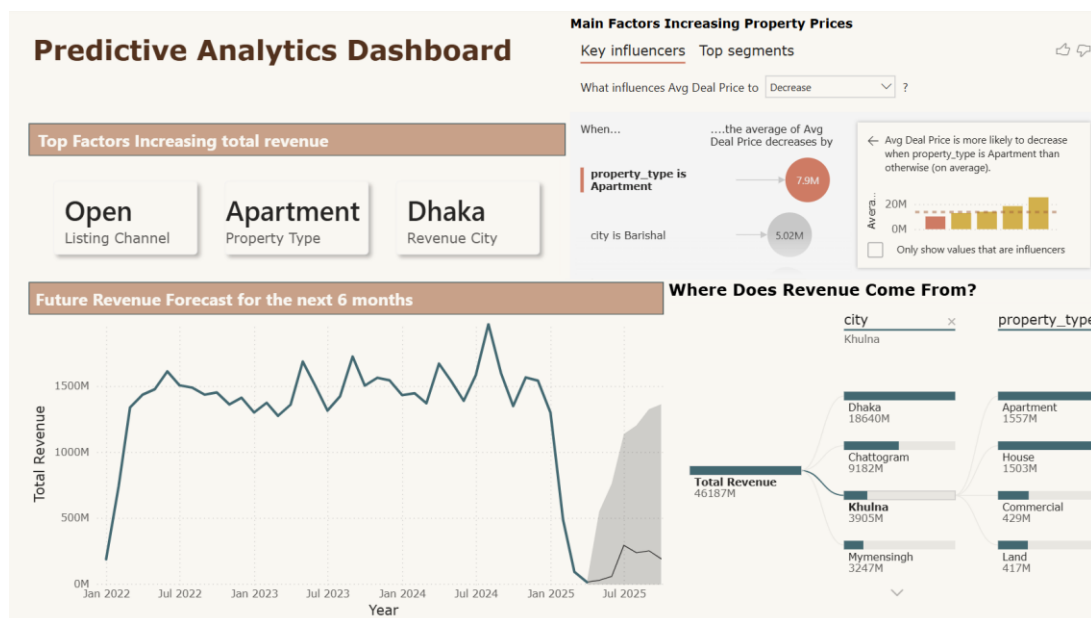


Figure 25 Predictive Analytics Dashboard

### 9.4.1 Top Factors Increasing Total Revenue

At the top of the page, three cards were used to highlight the main factors associated with higher total revenue, including the strongest listing channel, the top property type, and the highest revenue city. These cards provide a quick summary of the most important revenue drivers and help users immediately identify which categories contribute the most to business performance. This is useful because agents and management can focus more on the channels, property types, and cities that generate stronger revenue.

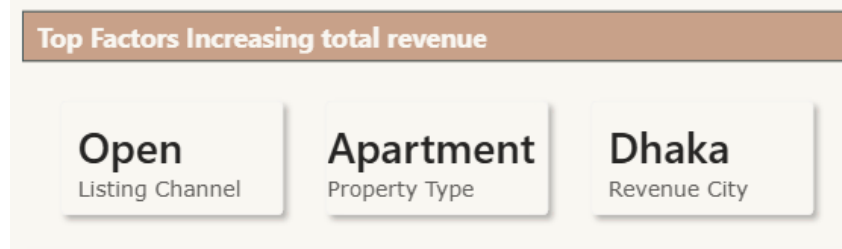


Figure 26 Top Factors Increasing Total Revenue

### 9.4.2 Future Revenue Forecast

A line chart with Power BI forecasting was used to predict future revenue for the next six months based on historical revenue trends. The actual revenue values are shown across previous months, while the forecasted area represents the expected future revenue. This visual supports the predictive analytics part of the dashboard because it uses past revenue patterns to estimate future values. It helps management prepare for possible increases or decreases in revenue and supports better planning for pricing, marketing, and sales strategies.

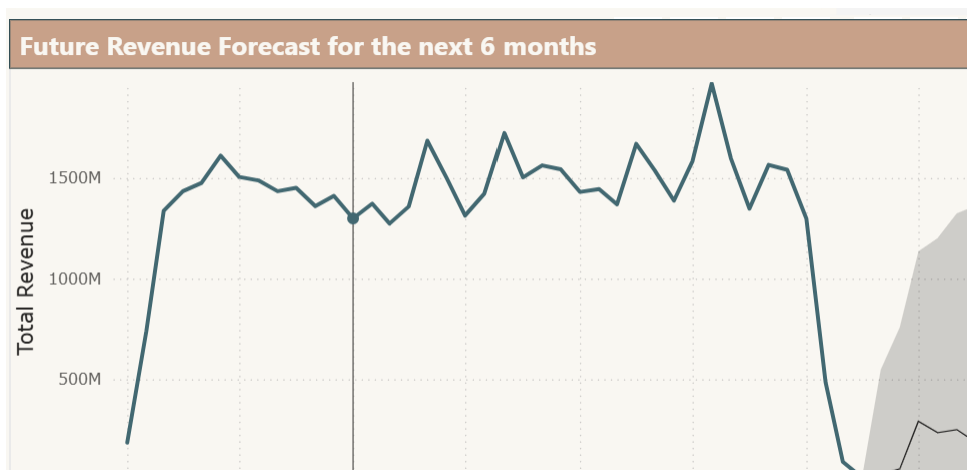


Figure 27 Future Revenue Forecast

### 9.4.3 Key Influencers

The Key Influencers visual was used to identify the main factors that affect the **average deal price**. This visual analyzes how different variables in the dataset influence the selected target value. In this dashboard, it helps explain which factors are related to higher or lower average deal prices, such as property type and city. This supports diagnostic analysis because it does not only show the result, but also helps explain why deal prices may change. Real estate agents can use this insight to understand which property features or locations are connected with higher deal values.

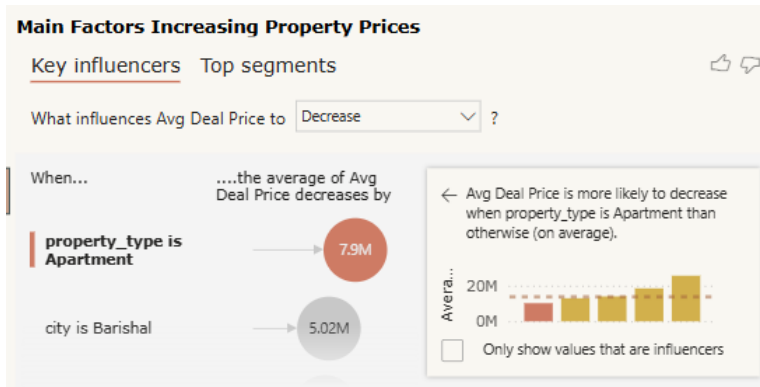


Figure 28 Key Influencers

### 9.4.4 Decomposition Tree

The Decomposition Tree was used to break down total revenue into smaller categories, such as city and property type. This visual allows users to explore revenue sources step by step and identify which categories contribute the most to total revenue. It is useful because it gives users an interactive way to investigate revenue performance instead of only viewing one summary number. For example, users can expand the tree to see which city generates more revenue and then continue analyzing which property type contributes most within that city.

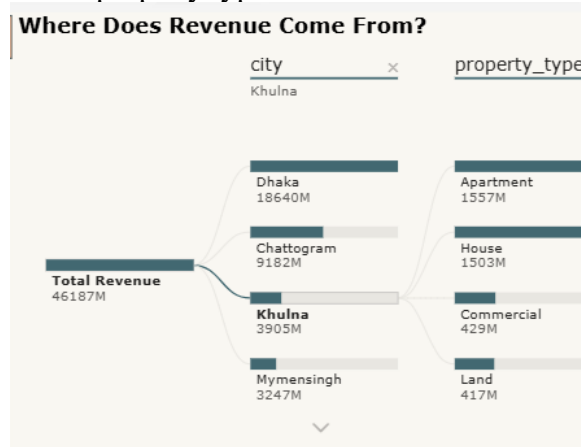


Figure 29 Decomposition Tree



### 9.5 Custom Calculations and Metrics

Several DAX measures were created to support the dashboard analysis. These measures were used in KPI cards and charts to summarize important business indicators related to revenue, transactions, listings, agents, and commission performance.

| Measure                   | Purpose                                                          | DAX Formula                                                       |  |
|---------------------------|------------------------------------------------------------------|-------------------------------------------------------------------|--|
| <b>Total Revenue</b>      | Calculates the total value of real estate deals.                 | Total Revenue =<br>SUM(transactions[deal_price])                  |  |
| <b>Average Deal Price</b> | Shows the average price of transactions.                         | Avg Deal Price =<br>AVERAGE(transactions[deal_price])             |  |
| <b>Total Commission</b>   | Calculates the total commission amount earned from transactions. | Total Commission =<br>SUM(transactions[commission_amount])        |  |
| <b>Total Deals</b>        | Counts the number of unique transactions.                        | Total Deals =<br>DISTINCTCOUNT(transactions[transaction_id])      |  |
| <b>Active Listings</b>    | Counts listing records in the listings table.                    | Active Listings =<br>COUNTROWS(listings)                          |  |
| <b>Total Customers</b>    | Counts unique customers.                                         | Total Customers =<br>DISTINCTCOUNT(customers[customer_id])        |  |
| <b>Total Cities</b>       | Counts unique customer cities.                                   | Total Cities =<br>DISTINCTCOUNT(customers[home_city])             |  |
| <b>Total Channels</b>     | Counts unique acquisition channels.                              | Total Channels =<br>DISTINCTCOUNT(customers[acquisition_channel]) |  |



|                               |                                                     |                                                                                                                                                                                            |      |
|-------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Avg Customers per City</b> | Calculates average customers per city.              | Avg Customers per City =<br>DIVIDE([Total Customers],<br>[Total Cities])                                                                                                                   |      |
| <b>Top Revenue City</b>       | Shows the city with the highest revenue.            | Top Revenue City =<br>MAXX(TOPN(1,VALUES(transactions[city]),[Total Revenue],<br>DESC), transactions[city])                                                                                |      |
| <b>Top Property Type</b>      | Shows the property type with the highest revenue.   | Top Property Type = VAR TopTable =TOPN(<br>1,SUMMARIZE('properties','properties'[property_type],<br>[Total Revenue]),[Revenue],DESC)<br>RETURN MAXX(TopTable, 'properties'[property_type]) | "Rev |
| <b>Top Listing Channel</b>    | Shows the listing channel with the highest revenue. | Top Listing Channel = MAXX(TOPN(<br>1,VALUES(listings[listing_channel]),[Total<br>Revenue],DESC),listings[listing_channel])                                                                |      |

Table 3 Measures Tables

Avg Deal Price = `AVERAGE(transactions[deal_price])`

Figure 30 Example of Measure

## 10. Power BI features used

Several Power BI features were used to build the dashboard and support the analysis.

Power Query Editor was used to prepare and transform the data. The preparation included checking data types, filtering missing values in important columns, renaming columns, reordering columns, and creating custom columns such as Market Speed Category, Listing Month, Deal Month, Revenue Category, and Commission Rate.

Data Model View was used to create relationships between the five tables. These relationships allowed visuals to use fields from different tables correctly.



DAX Measures were used to calculate important KPIs such as Total Revenue, Total Transactions, Total Customers, Average Deal Price, and Average Days on Market.

Different visuals were used to present the data clearly. Cards were used to display key numbers, bar and column charts were used to compare cities and segments, donut charts were used to show category distribution, and line charts were used for revenue trends and forecasting.

A waterfall chart was used to show total revenue contribution by agent city. A scatter chart was used to compare agent experience and rating. Key Influencers and Decomposition Tree were used for deeper analysis of deal price and revenue.

Slicers and filters were also used to make the dashboard interactive and user-friendly. These features helped users explore the data based on city, acquisition channel, and agent location.

Overall, the Power BI features used in this project helped create a clear, interactive, and useful dashboard for real estate market analysis.

## **11. Problems faced**

During the project, several problems were faced while preparing and building the Power BI dashboard.

First, the dataset contained multiple related tables, so it was important to understand the role of each table before creating the model. The team needed to identify the correct relationships between customers, agents, properties, listings, and transactions.

Second, some important fields contained missing values. These values had to be filtered carefully because they could affect dashboard accuracy, especially fields related to agent rating, transaction values, and listing information.

Third, creating custom columns required testing different formulas to make sure the categories were meaningful. For example, the Market Speed Category needed to classify listings into Fast Sale, Normal Sale, and Delayed Sale based on days on market.

Fourth, some visuals required specific measures and correct relationships to work properly. If a relationship was missing or inactive, the visual did not show the expected result. Therefore, the data model had to be checked multiple times.

Finally, formatting the dashboard took time because the pages needed to be clear, professional, and consistent. The team adjusted chart positions, colors, labels, and slicers to improve readability.



Overall, these problems helped the team understand Power BI better and improve the quality of the final dashboard.

## **12. Limitations**

Although the dashboard provides useful insights, there are some limitations.

First, the dataset is limited to the available records and does not represent all real estate markets. Therefore, the results should be understood as insights from this dataset only.

Second, the forecast is based on historical data patterns. It may not fully capture real-world changes such as economic conditions, new regulations, interest rates, or unexpected market events.

Third, some analysis depends on the quality of the available columns. For example, agent performance analysis depends on agent rating, experience years, and transaction data. If these values are incomplete or inaccurate, the analysis may be affected.

Fourth, the dashboard focuses on descriptive, diagnostic, predictive, and simple decision-support analysis using Power BI visuals. It does not include advanced machine learning models outside Power BI.

Finally, the dashboard does not include live data streaming at this stage.

The analysis is based on the imported dataset, so the dashboard needs to be refreshed when the data is updated.

## **13. Future work**

In the future, the project can be improved in several ways.

First, live or regularly updated real estate data can be connected to Power BI to make the dashboard more dynamic. This would help users monitor market changes in real time.

Second, more external factors can be added, such as interest rates, economic indicators, population growth, and location development projects. These factors may improve the quality of analysis and forecasting.

Third, more advanced predictive models can be added to predict property prices, expected selling time, or customer buying probability.

Fourth, more decision-support recommendations can be added in the future, such as suggested pricing actions, marketing focus, and agent assignment recommendations based on dashboard results.



Fifth, the dashboard can include more drill-through pages for detailed analysis of each city, property type, or agent. This would allow users to explore the data at a deeper level.

Finally, accessibility can be improved by adding more tooltips, clearer descriptions, and better visual contrast. This would make the dashboard easier to use for different types of users.

#### **14. Conclusions**

This project successfully developed an interactive Power BI dashboard for Real Estate Market Analysis. The dashboard combines data from customers, agents, properties, listings, and transactions to provide a clear view of market performance.

The project transformed raw real estate data into useful insights through KPIs, charts, filters, DAX measures, and predictive visuals. The Overview Dashboard summarized the main market indicators, while the Customers Analysis page explained customer distribution and acquisition channels. The Agent Performance page helped compare agent results, and the Predictive Analytics Dashboard supported future revenue forecasting and deeper factor analysis.

Overall, the dashboard supports better decision-making by helping users understand revenue trends, customer behavior, property performance, agent performance, listing speed, and future market direction. It also helps users identify which cities, property types, agents, and acquisition channels may need more attention based on the dashboard visuals.



## **Appendices**

### Appendix A: Read me file